

Reality Overlay Learning App (ROLA)

Updated Software Requirements and Development Roadmap

Module-wise Organized Version

This document combines the original SRS module list with the additional ROLA numbered prompts PDF, reorganized by module instead of prompt number.

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Document Sources

- Original SRS: Reality_Overlay_Learning_App_SRS.pdf
- Additional source: ROLA_numbered_prompts.pdf
- Organization rule: prompt labels and prompt numbering are removed; content is grouped by development module.

Project Summary

ROLA is an Android-based Augmented Reality educational application that lets learners scan real-world objects, recognize them with AI, and view learning information through AR experiences. The app combines Kotlin, Jetpack Compose, Material 3, ARCore, TensorFlow Lite, Room, Firebase, voice explanation, quiz, translation, personalization, wearable preparation, cloud services, and knowledge graph capabilities.

Core objectives include interactive real-world learning, science and engineering education support, reduced dependency on traditional learning methods, and a scalable AI-powered AR education platform.

Module-Wise Development Roadmap

Module 1: Android Project Setup and MVVM Architecture

Goal: Create the professional native Android foundation for ROLA.

- Use Kotlin, Android Studio, Jetpack Compose, Material 3, and Compose Navigation.
- Apply Clean Architecture with data, domain, presentation, DI, and utility layers.
- Prepare Hilt, Coroutines, Flow/StateFlow, Room, Retrofit, OkHttp, Coil, JUnit, and MockK dependencies.
- Set minimum SDK to Android 10 / API 29 and target the latest stable Android SDK.

Module 2: Firebase Authentication

Goal: Support secure learner account access.

- Allow users to register, log in, log out, and manage profile identity.
- Keep authentication state lifecycle-aware and compatible with MVVM.
- Prepare Firebase-backed user profile storage and secure access rules.

Module 3: ARCore Camera Scanner

Goal: Open the camera, start an AR session, detect real-world surfaces, and create anchors.

- Integrate Google ARCore and Sceneform/Filament-compatible rendering.
- Handle camera permission, unsupported devices, session failures, and lifecycle interruptions.
- Enable horizontal and vertical plane detection plus environmental lighting.
- Track anchors for future AR educational overlays.

Module 4: TensorFlow Lite Object Recognition

Goal: Recognize real-world objects from AR camera frames and return names with confidence scores.

- Load a .tflite model and labels from Android assets.
- Convert camera frames to images, resize, normalize, and prepare tensors.
- Run inference asynchronously with frame skipping to protect AR performance.
- Return recognition results containing object name, confidence, and timestamp.

Module 5: AR Floating Information Overlay

Goal: Display educational object information in AR space.

- Show object name, scientific information, description, uses, and facts.
- Attach information panels to AR anchors.
- Prepare readable overlay layout for mobile AR viewing.

Module 6: Room Database and Learning History

Goal: Persist scanned objects and learning progress locally.

- Store scan history with scan ID, user ID, object ID, timestamp, and progress metadata.
- Cache object knowledge for offline-first learning.
- Expose history data through repositories and Flow.

Module 7: Firebase Knowledge Database

Goal: Sync educational content and user progress with cloud storage.

- Use Firestore for users, objects, scan history, quiz results, translations, models, and analytics.
- Prepare secure synchronization between local Room data and Firebase.
- Keep write access controlled and verified educational content protected.

Module 8: Text-to-Speech Assistant

Goal: Provide voice-based explanations for object learning.

- Convert educational object descriptions into speech.
- Support playback controls and locale-aware voice output.
- Handle unavailable voices and fallback behavior.

Module 9: Quiz and Gamification

Goal: Generate quizzes and reward learning progress.

- Create quizzes based on scanned object content.
- Track scores, progress, achievements, and learning consistency.
- Support adaptive difficulty in later personalization modules.

Module 10: Testing and Deployment

Goal: Prepare production-quality testing and release workflows.

- Test AR scanner, object recognition, overlays, voice, history, quiz, and cloud sync.
- Prepare Android release configuration, CI/CD, signing, and Google Play deployment.
- Validate performance, security, usability, and Android 10+ compatibility.

Module 11: Complete App Integration and AI Orchestration

Goal: Connect scanner, recognition, knowledge retrieval, overlays, voice, history, quiz, tutor, translation, and recommendations into one learning flow.

- Coordinate AI agents and user context across modules.
- Route recognized objects through knowledge lookup, explanation, quiz, and progress systems.
- Keep integrations modular so each feature can evolve independently.

Module 12: AI Tutor Chatbot and 3D Object Visualization

Goal: Support follow-up learning conversations and interactive 3D object exploration.

- Provide an AI tutor grounded in app knowledge, scan history, and quiz progress.
- Prepare 3D GLB/GLTF object visualization through SceneView/Filament.
- Allow learners to move from object scan to explanation, tutor discussion, and 3D inspection.

Module 13: Real-Time Translation and Multilingual Learning System

Goal: Enable users to view and listen to educational object information in multiple languages.

- Create translation architecture under presentation/translation, data/translation, and domain/model.
- Translate object information, AI tutor responses, quiz content, and voice explanations.
- Support English, Bengali, Spanish, French, German, Chinese, and Arabic with easy expansion.
- Create Language model with languageCode, languageName, nativeName, and supportedVoice.
- Create Material 3 language selection UI with current and available languages.
- Integrate translated content with AR information overlay, Text-to-Speech, AI Tutor, and quizzes.
- Implement offline translation cache, local language packs, Room entities, and Firestore translation sync.
- Handle speech locale availability, fallbacks, cache usage, latency, network security, and tests.

Module 14: AR Glasses and Wearable Device Compatibility

Goal: Prepare ROLA for future AR glasses and wearable XR devices with hands-free immersive learning.

- Create wearable architecture under presentation/wearable, data/wearable, and domain/model.
- Design a hardware abstraction layer for AR glasses, smart glasses, XR headsets, and future wearables.
- Create WearableDevice model with device metadata, connection type, battery, display capability, and status.
- Prepare Bluetooth/WiFi discovery, connection, monitoring, and secure pairing states.
- Define data flow from wearable camera to object recognition, learning information, and AR display.
- Prepare voice-first navigation, hand tracking, eye tracking, spatial mapping, and depth sensor APIs.
- Optimize small field-of-view UI, audio output, low-power mode, and performance impact testing.

Module 15: AI Personal Learning Engine and Adaptive Education System

Goal: Analyze learner behavior and generate personalized recommendations, difficulty, and learning paths.

- Create adaptive learning architecture under presentation/adaptive, data/adaptive, and domain/model.
- Track scans, explored categories, quiz scores, learning time, searches, difficult concepts, language, and goals.
- Create LearningProfile with progress, weak areas, favorite categories, skill level, and recommendation fields.
- Detect strong topics, weak topics, knowledge gaps, learning preferences, and improvement trends.
- Generate personalized roadmaps, review suggestions, adaptive quizzes, and AI Tutor personalization.
- Add Room and Firebase support for learning profiles and recommendations while protecting user data.

Module 16: Advanced Computer Vision and Scene Understanding System

Goal: Upgrade recognition into scene-aware multi-object computer vision.

- Create advanced vision architecture under presentation/vision, data/vision, and domain/model.
- Support multiple object detection, tracking, classification, bounding boxes, and confidence scoring.
- Prepare EfficientDet Lite, MobileNet SSD, YOLO mobile, and optimized TensorFlow Lite detection models.
- Track object identity across frames and smooth AR overlay movement.
- Analyze scenes, object relationships, contextual safety information, and educational recommendations.
- Optimize with frame skipping, quantization, GPU acceleration preparation, background inference, and memory controls.
- Test lighting, fast movement, low-end devices, multiple objects, and AR tracking stability.

Module 17: Cloud AI Infrastructure and Scalable Backend Architecture

Goal: Transform ROLA into a scalable cloud-powered education platform.

- Create backend folders for authentication, user management, object knowledge, AI services, analytics, recommendation, and storage.
- Design Mobile App -> API Gateway -> Backend Services -> Cloud Database -> AI Services architecture.
- Implement User, Object Knowledge, AI, Analytics, Storage, and Admin management services.
- Use Firebase Authentication, Firestore, Storage, Cloud Functions, rules, indexes, and sync preparation.
- Store object images, 3D models, AI model files, educational resources, and content versions securely.
- Prepare REST/GraphQL APIs for users, objects, search, recognition, chat, translate, recommendations, and analytics.
- Add role-based access control, encryption, monitoring, performance testing, and large-user scalability preparation.

Module 18: AI Knowledge Graph and Semantic Learning Network

Goal: Build a semantic knowledge system that understands relationships among objects, concepts, materials, and applications.

- Create knowledge graph data layer with repository, graph database manager, relationship engine, and semantic search.
- Create KnowledgeNode, KnowledgeRelation, ConceptGraph, and LearningPath domain models.
- Support node types such as Object, Material, Scientific Concept, Process, Application, Theory, and Technology.
- Model relationships such as is_a, made_of, used_for, related_to, part_of, causes, discovered_by, and applied_in.
- Enable semantic search and grounded AI Tutor explanations based on graph retrieval.
- Integrate Search, Adaptive Learning, Firebase-compatible graph caching, admin expansion, and verified content controls.
- Test relationship accuracy, graph navigation, semantic search, tutor responses, and recommendations.

Functional Requirements Mapping

Requirement	Mapped Modules
User Authentication	Module 2, Module 17
AR Camera Scanning	Module 3, Module 14, Module 16
Object Recognition	Module 4, Module 16
AR Information Display	Module 5, Module 13, Module 14
Voice Explanation	Module 8, Module 13, Module 14
Learning History	Module 6, Module 15, Module 17
Search System	Module 7, Module 18
Quiz System	Module 9, Module 15
Cloud Scale and Analytics	Module 15, Module 17

Non-Functional Priorities

- Performance: object recognition response should remain under three seconds and AR rendering should stay smooth.
- Security: authentication, user data, API communication, keys, and cloud rules must remain protected.
- Usability: scanner, overlays, translation, voice, tutor, quiz, and dashboards should remain simple for learners.
- Compatibility: Android 10+ and ARCore-supported devices are the baseline, with wearable preparation added later.